Unit no. ° 14

PROBABILITY: CONDITIONING AND WEIGHTED TREES

26 April to 24 May 2027

I Sample space, outcomes and events

Vocabulary

A random experiment has several possible outcomes, and we do not know in advance which one will be obtained. An outcome is a possible result; the sample space is the set of all outcomes. An event is a subset of the sample space; an elementary event contains a single outcome. The impossible event is $\varnothing$; the sure event is Omega.

Operations on events

$\overline{A}$ is the complementary event of A. $A \cap B$ means "A and B"; $A \cup B$, "A or B", possibly both. Two events are mutually exclusive if $A \cap B = \varnothing$; otherwise, they are not disjoint.

Example 14.1

A six-sided die is rolled. Write ?, A : "obtain an even number", B : "obtain 4", then $\overline{A}$, $A \cap B$ and $A \cup B$. A, B : are they compatible?

Solution:

Example 14.2

With $A = \{a; b; c; d; e\}$ and $B = \{b; e; f; g; h\}$, give the intersection and the union. Can we determine $\overline{A}$ without specifying the sample space?

Solution:

II Probability model and equally likely outcomes

Probability in a finite universe

A model assigns to each outcome a probability that is positive or zero; the sum is 1. The probability of an event is the sum of the probabilities of its outcomes. We have

$$P(\Omega) = 1, \quad P(\varnothing) = 0, \quad P(\overline{A}) = 1 - P(A).$$

With equally likely outcomes, all outcomes have the same probability and

$$P(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)}.$$

Equally likely outcomes are an assumption of the model, not a consequence of the list of outcomes alone.

Example 14.3

Assume the die is fair. For A = "even" and C = "greater than or equal to 5", calculate $P(A)$, $P(C)$, $P(A \cap C)$ and $P(A \cup C)$.

Solution:

Example 14.4

A drawing pin can land point up or point down. Can we deduce that each outcome has probability 1 /2 ? How can we propose a model?

Solution:

III Conditional probability from a table

Definition

For $P(A) > 0$, the probability of B given A is

$$P_A(B) = \frac{P(A \cap B)}{P(A)}, \qquad P(A \cap B) = P(A)P_A(B).$$

In an equally likely draw from a finite population:

$$P_A(B) = \frac{\text{Card}(A \cap B)}{\text{Card}(A)}.$$

We also write $P(B \mid A)$. $P_A(B)$ and $P_B(A)$ are generally not equal.

Example 14.5

A pupil is drawn uniformly at random from the class of 32 pupils studied earlier. G : "boy", N : "takes NSI".

	G	$\overline{G}$	Total
N	12	9	21
$\overline{N}$	6	5	11
Total	18	14	32

Calculate $P(G)$, $P(N)$, $P(G \cap N)$, $P_N(G)$ and $P_G(N)$.

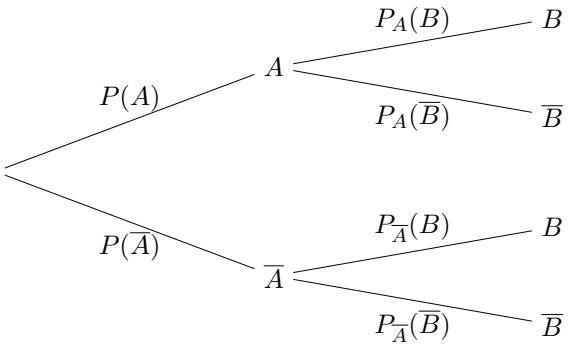
Solution:

Watch out. Conditioning requires the known event to have non-zero probability; we do not divide by zero.

IV Weighted trees and the probability of a path

Rules

At each node, the sum of the probabilities on the outgoing branches is 1. A branch of the second level carries a conditional probability. A path represents an intersection of events; its probability is the product of the probabilities written on this path.



Example 14.6

We know $P(A) = 0{,}6$, $P_A(B) = 0{,}7$ and $P_{\overline{A}}(\overline{B}) = 0{,}9$. Complete the other branches and calculate the probabilities of the four paths.

Solution:

Example 14.7

Justify multiplying along the path A then B from the definition of $P_A(B)$.

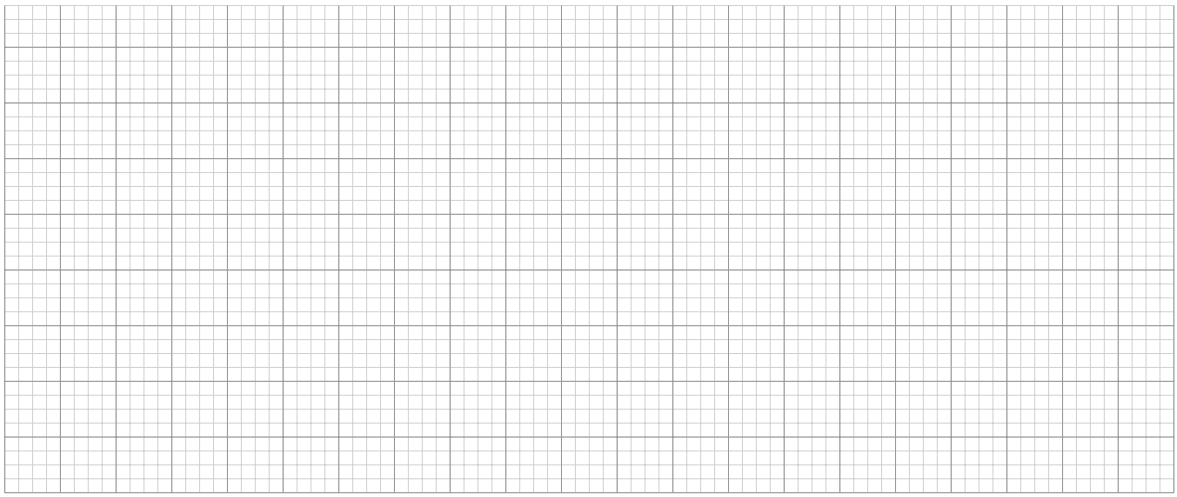
Solution:

V Moving from a table to a tree

Example 14.8

For the class of 32 pupils, draw a tree starting with NSI / no NSI, then boy / girl.
Then construct the tree in the reverse order. Check, in both trees, the probability of choosing a girl given NSI.

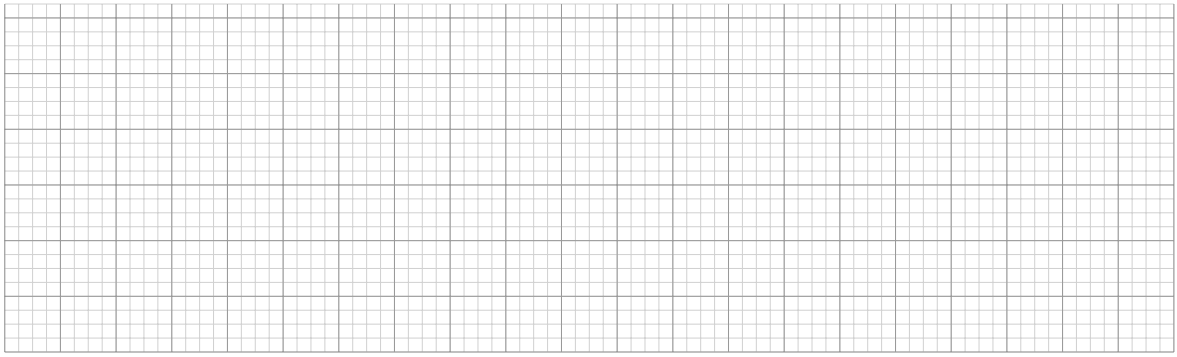
Solution:



Example 14.9

An urn contains 3 red balls and 2 blue ones. We draw two balls in succession **without replacement**. Construct the colour tree and calculate the probability of drawing two reds.

Solution:



VI Reversing the conditioning and false positives

Example 14.10: Fictional component inspection

A batch contains 10000 components, 100 of which are defective. A test is positive for 99 of the defective components and for 495 of the non-defective components. A component is chosen uniformly at random. D : " defective"; T: "positive test".

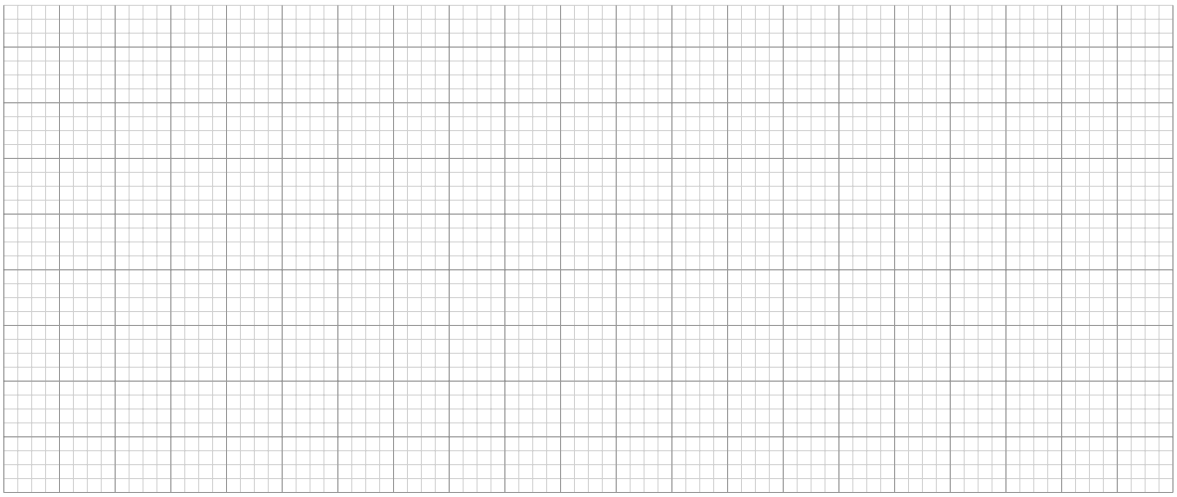
1. Complete the two-way table: test / true status.

2. Calculate $P_D(T)$ and $P_T(D)$.

3. Give the number of false positives and false negatives.

4. Calculate sensitivity $P_D(T)$ and specificity $P_{\overline{D}}(\overline{T})$.

Solution:



Read the test result

"Positive among the defective" and "defective among the positives" are two different conditionings. Always name the reference group before calculating.

VII Simulation and law of large numbers

Simplified version

When an experiment is repeated a large number of times in the same model with independent repetitions, the observed frequency of an event is, except in rare cases, close to its probability.
It fluctuates: it does not necessarily get closer to it with each new experiment.

```
from random import Random

def frequences_six(n, graine=1):
    # n integer >= 1 ; model of independent tosses
    rng = Random(graine)
    succes = 0
    frequences = []
    for k in range(1, n+1):
        if rng.randint(1, 6) == 6:
            succes = succes+1
        frequences.append(succes/k)
    return frequences
```

Example 14.11

Plot the frequencies for $n = 1000$, with several seeds. Compare with $1/6$. Why fix the seed?
Can we require the final frequency to be exactly $1/6$?

Solution:

Example 14.12

Modify the program to track the frequency of the event "even result". What reference probability to use?

Solution: